The development of velum coordination in children: a real-time MRI study

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Abstract

This study examines the development of velum coordination in American English speaking children using real-time MRI. We focus specifically on three aspects of velum coordination —timing, displacement, and duration, as a function of the syllable positions of the nasal, i.e. onset, coda, or juncture geminate positions. It is found that the two oldest speakers, aged 15, generally show clear distinctions along these dimensions depending on the syllable position of the nasal and that they pattern closely with findings from adults. The younger speakers, ages 5-14, do not consistently distinguish nasals by syllable position along these dimensions. For timing, all younger speakers regularly employed very early velum lowering. These findings are discussed with respect to previous theories on the development of coordination and coupling modes in speech production, particularly within complex segments.

1 Introduction

Speech production necessitates the coordination of various articulators to form constrictions in the vocal tract. Mastering this coordination is a fundamental challenge in speech development, particularly given that the child learner must cope with a growing vocal tract (McGowan et al., 2004). Crucially, articulatory coordination entails coordination in both *space* and *time*, and while spatiotemporal aspects of coordination, namely coordination or gestural overlap, have received considerable attention in prior work on speech development (Rubertus and Noiray, 2018; Cychosz et al., 2021), articulatory timing is notably less well researched. The current study aims to address this considerable gap by looking at articulatory timing specifically, among other aspects of articulatory coordination.

Coordination within segments, such as in complex segments, often requires coordinating multiple constrictions in a stable manner that is distinct from coordinating across segments (Shaw et al., 2021). Nasal consonants are a prime example of this as they require both the lowering of the velum and the raising of the tongue tip (TT). Previous work on adults has demonstrated that the timing of these two gestures can vary cross-linguistically (Proctor et al., 2012), depending on the syllable position of the nasal (Krakow, 1999; Byrd et al., 2009) and across prosodic conditions (Proctor et al., 2012). The development of coordination in complex segments such as nasals is severely understudied.

The current study investigates velum coordination in American English speaking children using real-time MRI (rtMRI). This study represents one of the first studies to investigate articulatory timing in children, particularly by means of rtMRI. Nasals are a viable case study in examining articulatory coordination given that they involve both a velum lowering gesture and tongue tip raising gesture. While clear adult patterns in velum coordination have been established, how these patterns develop in children is largely a mystery. Indeed, there have been studies on nasal coarticulation in children using acoustic measures, but this does not afford the same degree of clarity offered by rtMRI in illuminating how children coordinate the velum in speech.

The rest of the paper is organized as follows. Section 1 dives deeper into the previous literature on articulatory coordination, coupling modes, and speech development with reference to nasals specifically and states the research questions and hypotheses of the current study. Section 2 outlines the methodology employed in the current study, giving an overview of the equipment and analysis used. Section 3 discusses the results of the study, dividing them into results concerning velum timing, displacement, duration, and the interaction between the three. Section 4 discusses the implications of the result with respect to theories of articulatory coordination, coupling modes, and development, limitations of the current study, and ways to extend the current work.

1.1 Articulatory coordination in nasals & complex segments

Nasal consonants require both the formation of an oral constriction and a lowering of the velum to allow airflow through the nasal cavity. Unlike other articulators that typically involve muscular *activation* to initiate movement towards its speech production goal, it is primarily *inactivation* of the levator veli palatini that results in velum lowering during nasals (Bell-Berti, 1993; Zajac, 2015). Nasal breathing with a lowered velum position is also typical during rest. Thus, gestural coordination in nasal consonants entails coordinating inactivation of the levator veli palatini and activation of the lingual and mandibular musculature involved in the formation of a TT constriction. We postulate that this facts plays a critical role in the development of velum coordination in nasal consonants.

The timing of the oral and velum gestures has been found to vary cross-linguistically (Proctor et al., 2012; Krakow, 1999), depending on the syllable position of the nasal (Krakow, 1989, 1999; Byrd et al., 2009; Oh, 2021), and across prosodic conditions (Byrd et al., 2009; Proctor et al., 2012). In a comparison of French and English, Proctor et al. (2012) found velum-TT timing to be different between French nasal vowels and consonants, with velum loweing preceding vowel target achievement in nasal vowels, a pattern that English patterns more closely with. Lag between velum and TT onset has been found to vary between onset and coda nasals in American English (Krakow, 1989, 1999; Byrd et al., 2009), but not so for Korean nasals, with syllable position effects impacting the lag between velum raising and TT onset instead (Oh, 2021).

Focusing on American English, nasals have been shown to exhibit clear timing patterns in adult speech dependent on the syllable position of the nasal. In onsets, near synchronous timing of the velum lowering and oral constriction gestures is observed, while for coda nasals the two gestures asynchronously, with the velum lowering being initiated before the formation of the oral constriction (Krakow, 1989, 1999; Byrd et al., 2009). The earliest study demonstrating this was Krakow (1989), who employed a Velotrace to track the position of the velum during nasal sequences. Krakow (1989) found that, during the production of /m/, lowering of the velum occurred well before the labial constriction in coda /m/ and that for onset /m/ the timing was much more synchronous. These results were later replicated in Byrd et al. (2009) using rtMRI and focused on the coronal nasal /n/. These timing distinctions can be represented graphically using a *gestural score*, a gestural activation matrix that has vocal tract organs on the y-axis and time on the x-axis, shown in Figure 1 for the words "nab" (left) and "ban" (right). Note that in "nab" the onset if the gestures for the velum, TT, and tongue body all occur at the same time. In "ban", it can be seen that the onset of the velum gesture occurs well before that of the TT, while the vowel gesture occurs before the velum gesture.

In addition to the bimodal distribution in timing, there is also evidence that this bimodality extends also to magnitude of the velum lowering. Generally, a lower velum position is found in coda nasals (Krakow, 1999; Byrd et al., 2009; Blaylock et al., 2016). This has also been found in other languages, e.g. Hindi (Henderson, 1984). Duration of the velum lowering gesture has also been found to vary as a function of syllable position, with coda nasals having longer velum gestures than onset nasals (**?**). In all, previous work has established that speakers of American English modulate velum gestures in terms of timing, magnitude, and duration dependent on the syllable position of the nasal.

Articulatory coordination in nasals is also relevant to previous work on the stability of coordination in complex segments, i.e. segments involving multiple constrictions. It has been proposed that the crucial distinction between complex segments and segment sequences is the stability of

their coordination (Byrd, 1996; Shaw et al., 2021). For example, Shaw et al. (2021) found that onset lag between the two gestures in palatalized consonants in Russian did not vary considerably as a function of varying gestural duration, while similar labial-palatal sequences in English did. In Hausa, Oh et al. (2024) discovered that stability in onset lag between the larynx and oral gestures is critical in distinguishing implosives and their pulmonic counterparts, with implosives displaying greater stability in this dimension. Similarly, Korean nasals have been found to have greater stability in the lag between velum target achievement and TT raising than onset lag between the two gestures, suggesting the former is part of goal in Korean nasal production (Oh, 2021). While timing patterns have been investigated in American English nasals, the stability of these patterns has not, and as such, it is unknown what temporal relation between the velum and TT exhibits stability in these segments.

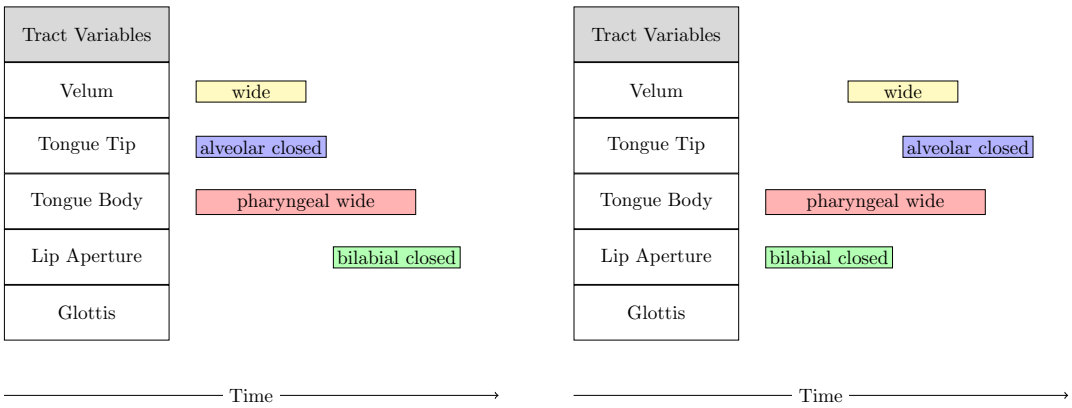

Figure 1: Gestural scores for word 'nab' (left) and 'ban' (right).

1.2 Development of articulatory coordination & nasals

A particularly robust findings from studies on the development of articulatory coordination is that children are more variable or less stable in their coordination than adults (Green et al., 2000; Smith and Zelaznik, 2004; Green and Nip, 2010; Terband et al., 2011). In one of the most extensive studies to date, Smith and Zelaznik (2004) looked at variability in lip aperture across 180 children and adults using a point tracking based method. They found the general trend that very young children exhibited much higher variability than adults, but that even kids aged 14 show considerably more variability than young adults. A particularly striking additional finding was the plateau in variability roughly between the ages 6-12, where little to no change in variability was observed across subjects. In some cases, the subjects aged 12 showed *more* variability than their much younger counterparts. There is also a great deal of evidence that children coarticulate more than adults (**???**Abakarova et al., 2022), though this finding is not always replicated (Noiray et al., 2013).

A number of studies have investigated velopharyngeal function and nasal coarticulation in children (Flege, 1988; Ha and Kuehn, 2006). Acoustic-perceptual studies have suggested that some of the early infant vocalizations are produced with a lowered velum and that infants are nasal breathers (Zajac, 2015). This likely stems from the close proximity of the epiglottis to the velum in infancy, which later separate as the larynx descends (Zajac, 2015). Changes in the velopharyngeal region during development can also impact velum action during speech. Lymphoid tissue in the nasopharynx, termed the adenoids, grow on the rear pharyngeal wall around age 2 and have been speculated to impact velum closure and levator veli palatini activation, given that for some children velum closure may involve velar-adenoidal contact rather than pharyngeal contact (Zajac, 2015; Perry et al., 2017). In early adolescence, atrophy of the adenoids forces adjustment of the velum raising gesture, given changes to the posterior region with which velum contact may previously have been made (Zajac, 2015).

While one early study on anticipatory nasal coarticulation in children found that children have less nasalization than adults (), more studies since then have found either no difference between

adults and children or that children have more nasalization (Flege, 1988; Leeper et al., 1998; Ha and Kuehn, 2006). Flege (1988) investigated anticipatory and carryover nasalization on the vowel in d_n and n_d contexts. They found that children between the ages 5-10 did not differ significantly in their vowel nasalization from adults in these two contexts. The authors go on to suggest that early velum lowering in final nasals may require less effort than more rapid velum lowering closer to the onset of the oral constriction. Ease of effort was also argued to explain earlier velum lowering in children with cleft palate (Ha et al., 2004; Ha and Kuehn, 2006), with the authors going on to suggest that perhaps less nasalization is associated with the development of speech motor control. A crucial shortcoming of this previous work is that it has ignored the larger context concerning velum timing patterns employed in adult speech. Specifically, this work has not considered variations in velum timing due to different syllable positions of the nasal. Often the focus has been on anticipatory effects, i.e. coda nasals only, or on nasals in VCV sequences with ambiguous syllabification. For example, while Flege (1988) did include onset and coda nasal conditions, they only looked at nasalization on vowels in the same word as the nasal, so any potential effects preceding the vowel in the n_d context were ignored.

To our knowledge, no previous study has investigated articulatory timing in children. While some studies have carried out acoustic experiments that can be used to infer the timing of articulatory actions (Flege, 1988; Ha and Kuehn, 2006; Howson and Redford, 2021), no study has actually quantified the timing of one articulatory event relative to another with articulatory data from children. The current study is then the first study to investigate articulatory timing in children, and one of the first studies to employ rtMRI to study child speech production.

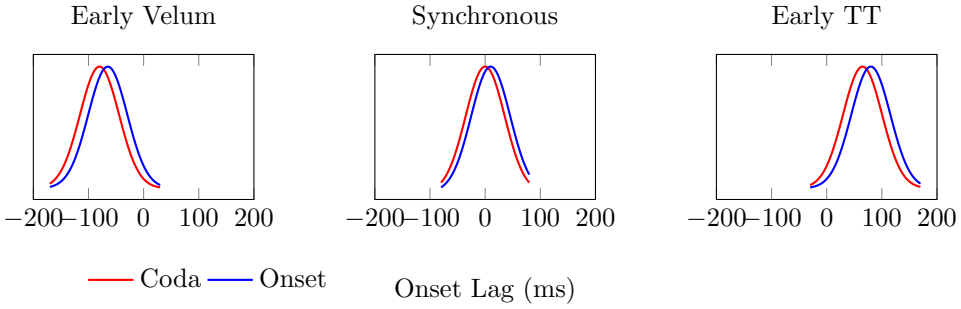

Figure 2: Three hypothetical starting points for the development of velum-TT timing as a function of the syllable position of the nasal. Negative values indicate the velum precedes the TT, zero indicates perfect synchrony, and positive values indicate the TT precedes the velum.

1.3 Research questions & predictions

In this study, we investigated three primary research questions regarding the development of velum coordination in American English nasals:

1. What velum timing strategy is employed by child speakers of American English in the production of nasals?

2. How do child speakers of American English modulate velum gesture magnitude and duration as a function of the syllable position of the nasal?

3. What timing relation between the TT and velum is stable in American English nasals and at what age do children exhibit this stability?

In regard to Q1, we make the basic assumption that younger children start out with a single timing strategy. That is, rather than exhibiting the bimodal timing pattern of adults, the child timing distributions are unimodal, regardless of the syllable position of the nasal. This follows from the evidence showing that the timing patterns are language-specific and must therefore be learned through experience (Cychosz et al., 2021). This premise does not necessarily assume that children

are coordinating the velum and TT, given that it is possible that they start out not coordinating the two gestures. The question of stability (Q3) addresses this. Following the uniform timing assumption, theoretically there are three possible timing patterns, visualized in Figure 2. These include early velum lowering, synchronous timing, or early raising of the TT. We rule out early TT raising given that such a timing pattern is not present in the input and that no previous work motivates this possibility. This leaves two competing hypotheses: early velum lowering or synchronous timing.

The synchronous timing hypothesis has been predicted by work within the coupled-oscillator framework. Specifically, it has been suggested that the in-phase mode (synchronous timing) is more readily available and more stable and that children have learned this mode during babbling, given the universal frequency of CV syllables (Goldstein et al., 2006; Nam et al., 2009). While these predictions were primarily made with respect to CV timing and coordination, the same coupling modes have been proposed to underlie coordination in complex segments (Byrd, 1996; Byrd et al., 2009). Kent (1983) also suggested that child speech is more likely to exhibit synchronous timing patterns than adults. Both propositions would suggest that children start out producing all complex segments with near synchronous timing among their respective gestures and that asynchronous timing emerges later in codas. This view is also supported by work showing that children acquire coda complex segments later than onsets (Lin and Demuth, 2015).

The early velum lowering hypothesis can be motivated by limited previous experimental work and aspects of velopharyngeal function. As noted above, both Ha and Kuehn (2006) and Flege (1988) found that children ages 4-10 exhibited adult-like degrees of vowel nasalization preceding coda nasals. This would indicate early velum lowering for coda nasals and, given our assumption of uniform timing, hints at early velum lowering as the strategy employed by children. One must also consider that velum lowering entails *inactivation* of the levator veli palatini and that the resting state of the velum is in its lowered position (Bell-Berti, 1993; Zajac, 2015). Given knowledge of the upcoming nasal, to time the velum to be near synchronous with the TT gesture requires inhibition of velum lowering. From this perspective, lowering the velum early may simply be easier for the child learner than maintaining *activation* of the levator veli palatini and coordinating its relaxation with TT raising. Similarly, it has been argued that earlier velum lowering in children with cleft palate may be due to effort minimization (Ha and Kuehn, 2006). The role of inhibition was also proposed by Howson and Redford (2021) for the acquisition of articulatory coordination in liquids. They argue that children produce early tongue body retraction in /l/ and /r/ due to a lack of inhibition of anticipatory tongue body constrictions for upcoming vowels.

For Q2, recall above that previous work has demonstrated that adults also show a bimodal pattern in velum magnitude and duration, dependent on the syllable position of the nasal. Onset nasals have shorter durations and lesser magnitude than coda nasals. Similar with timing, we assume that this pattern is language-specific and must be learned. Therefore, as with timing, is it predicted that younger children do not make distinctions in velum magnitude or duration when varying the nasal syllable position, but rather they adopt a global strategy that is employed across the board.

Q3 addresses not only a question of development but also of timing stability in American English nasals more generally. Although previous work has established timing patterns in adult productions on American English nasals, they did not investigate the stability of these patterns (Krakow, 1989, 1999; Byrd et al., 2009). Therefore, it is unknown what sort of timing relation is stable among the velum and TT in the production of nasals for this language. Work on palatalized consonants in Russian showed onset lag to be stable (Shaw et al., 2021), while for Korean nasals it was shown that velum target to TT onset lag was more stable than onset lag (Oh, 2021). In terms of developmental patterns, given the robust evidence that children are more variable in their speech motor skills (Green et al., 2000; Smith and Zelaznik, 2004), we predict that younger speakers will display less stability in the timing relations between the velum and TT. This also leaves open the possibility that younger speakers are not yet coordinating the velum and TT, but rather implementing an independent velum timing strategy. Concretely, this could involve early velum lowering or near synchronous timing on the one hand, but variability in the velum-TT timing relationship as a function of varying velum magnitude or duration on the other.

2 Method

2.1 Subjects

A total of seven subjects took part in the study, all of whom were female. Subject ages can be seen in Table 1 and range from 5-15 years old. All subjects were monolingual American English speakers and reported no history of hearing or speech disorders. The subjects include three pairs of siblings and efforts were made to have each sibling complete the study independently. Specifically, the non-participating sibling wore noise-canceling headphones while the participating sibling completed the familiarization task and MRI scan. All subjects were recruited from the greater Los Angeles area and IRB approval was granted for the study. One speaker was excluded from the study due to two reasons. First, the speaker had silver crowns which produced an artifact in the MRI video that greatly obscured the TT and lowered the resolution of the rest of the image, making analysis difficult. Second, the speaker had considerable head movement which also made analysis difficult as it relied on fixed region-based analysis that is only accurate if head movement is minimal.

ID	Age	Sex
Spk5	5	F
Spk8	8	F
Spk11a	11	F
Spk11b	11	F
Spk14	14	F
Spk15a	15	F
Spk15b	15	F

Table 1: Subjects in the current study with their age and sex. Subject IDs assigned based on age.

2.2 Stimuli

The stimuli used in the study were adapted from Byrd et al. (2009) (Table 1). Given that Byrd et al. (2009) had adult subjects, this allows for direct comparison of the results with child subjects to adults presented with similar stimuli. Specifically, the stimuli in Byrd et al. (2009) consisted of singleton nasals in onset, coda or juncture geminate position within a two-word phrase. The onset and coda positions are the primary comparison being made for velum timing, while the juncture geminate items create another context, with the possibility of the geminate patterning with either onset or coda nasals. In addition to the syllable conditions, labial and alveolar frames are used to vary the surrounding context of the target nasals. In the both cases, the nasal is surrounded by the vowel [oʊ] and then either a labial or alveolar. The one exception to this is the items where the second word is "OK", which would have a velar to the right. The use of the vowel [oʊ] is to induce jaw lowering and mouth opening, so as to more easily identify the oral constrictions during the target nasals. In total, there are 6 target items.

Stimuli were presented with pictures that match the target phrase. For example, the pictures for the phrase "bow no" can be seen in Figure 3. Each set of pictures was chosen so as to be age appropriate and all were taken from open source image websites. Each target item was repeated ten times, for a total of 60 spoken items. These items were randomized and blocked within singleton/cluster groups. Subjects were familiarized with the six pictures and the carrier phrases beforehand. The target words were elicited using the pictures and subjects were instructed to say the target word if they said a different word.

2.3 Data collection

The vocal tract of the speakers was imaged in midsagittal orientation using a 0.55T MRI scanner with a custom upper airway receiver coil (Muñoz et al., 2023), with a reconstructed frame rate of 99 frames/sec (See Kumar et al. (2024) for details). Audio was collected at 16 kHz. All data was acquired at the USC Dynamic Imaging Science Center. Subjects lied prone within the MRI

Figure 3: Pictures used for the target phrase "bow no". Subjects were shown this image and were also cued to repeat after the experimenter who produced the target phrase in a carrier phrase. For these pictures, the subject would hear and repeat "I can type bow no for you".

Syllable Position	**Labial frame**	**Alveolar frame**
	I can type ___ for you	I can write ___ to you
onset	bow no	toe note
coda	bone oh	tone oat
geminate	bone no	tone note

Table 2: Target phrases containing singleton nasals where labial and alveolar frames surround the target nasal with either a labial or alveolar consonant.

scanner with their head placed within the speech coil. Headphones were worn by all subjects to lessen the noise from the scanner and these also acted as headphones through which they could hear the experimenter. The picture stimuli were presented to subjects via a television placed behind the scanner and a mirror was placed above the subjects face to allow them to view the television. It was confirmed for each subject that they had a full view of the television through the mirror and adjustments were made to ensure this.

The general procedure involved two parts: 1) familiarization of the stimuli; 2) the MRI scan. During the familiarization task participants were all shown the pictures to be presented during the scan and the target phrases were elicited. In cases where the speaker produces a word that diverges from the target, they were instructed to produce the target when seeing a particular image. All ten images were gone through with participants at least twice. Following presentation of the images, the participants were introduced to the repeat-after-me paradigm and were instructed to repeat after the experimenter and to use the images as cues to the words spoken. Each subject was instructed through multiple examples before being tested on several further examples without instruction.

Each scan began with both an audio test, to ensure the subject's audio volume is sufficient to be captured by the microphone, and a hearing test to ensure subjects could hear the experimenter directions and cues. Each scan lasted roughly 30 seconds and consisted of four target sentences, giving a total of 15 scans per subject for the full stimuli with 10 repetitions. In cases where subjects made errors, only errors within the target phrase itself resulted in a repetition of that item. Errors in the carrier phrase, e.g. substituting "to" for "for" or vice versa, did not result in a repetition of that item.

2.4 Data Analysis

2.4.1 Tracking velum and TT kinematics

Gestural landmarks for the velum lowering and oral constriction formation (onset, target, and release) were determined using region-of-interest (ROI) analysis. ROI analysis is commonly used in image analysis methods for rtMRI and has been used in a range of studies (Lammert et al., 2010; Proctor et al., 2011; Oh et al., 2021; Lu et al., 2022). Velum lowering movement was analyzed using the approach detailed in Carignan et al. (2021). First, triangular vocal tract regions-of-interest (ROI) were defined for the velum per-speaker, using both anatomical landmarks and observation of the resting state during nasal breathing (See Figure **??**). The use of triangular ROIs allows for minimizing intrusion of the tongue (Oh et al., 2021; Oh, 2021). The horizontal dimension stretches from the edge of the posterior nasal spine to the rear pharyngeal wall or the edge of the adenoid glands[1]. The vertical length was defined based on the lowest point of the velum during rest.

Pixels within these defined regions across all files were submitted to a principle component analysis (PCA). From this trained PCA model, the loadings from the first principle component (PC1) for each frame generate a 1D timeseries of velum lowering/raising with a sampling rate of 99 Hz (See Figure 5). Given that variation in velum movement primarily exists along dimensions of lowering/raising, PC1 can be said to be the component that best captures this dimension of variation. On average, PC1 captures % of variance across speakers, with a standard deviation of _. Following Carignan et al. (2021), PC1 loadings were scaled within (0, 1) based on the maximum and minimum loading across all files per speaker. The loadings then represent a proportion the total range of velum movement. Given that the sign of the PC1 loadings is arbitrary per-speaker, the sign of the loadings was flipped so that positive values correspond to greater velum lowering. This was determined manually per-speaker by inspecting the timeseries in MVIEWRT (Proctor et al., 2011).

ROIs were also defined independently per-speaker for the TT. Circular ROIs were defined by observing TT constrictions during the target nasals, such that the anterior edge contains the area of the alveolar ridge where such constrictions are made and the posterior edge contains the left-most edge of the tongue during adjacent vowel constrictions. The size of the region was defined such that a portion of the tongue body is always present within the region, based on observation of TT gestures across multiple files. Change in average pixel intensity was used to track TT kinematics (**??**), such that higher average pixel intensity is indicative of greater constriction, i.e. TT raising. For all speakers, excluding Sk5 and Spk7, a single ROI was defined for each articulator across all files. Two regions were used for Speaker 5 and Speaker 7 due to head re-positioning mid-study requiring new regions be defined.

All velum and TT ROIs were defined using a publicly available Matlab toolbox[2]. The resulting timeseries from the velum ROI were smoothed using LOESS, with a span of 30 data points, while for the TT ROI timeseries locally-weighted linear regression was used ($H = 0.05$). From 1D timeseries for each ROI, gestural landmarks were defined using a custom Python script based on the `find_gest` algorithm described in (Tiede, 2010). The relevant TT and velum gestures for each target phrase were selected automatically within each file using phrase-level alignments from TextGrids. First-pass alignments were automatically generated using the Montreal Forced Aligner (McAuliffe et al., 2017) and corrected by hand.

Movement onset (GONS) was defined to be the point at which velocity reached 20% of its first maximum velocity (PVEL) for the movement towards the target. Target (NONS) was identified as the time-point after maximum velocity and before maximum extremum position at which velocity fell below a 20% threshold of the first maximum velocity. The maximum displacement of the gesture was defined at the velocity minimum closest to the movement's peak (MAXC). The target offset (NOFFS) was calculated at the point following maximum displacement at which velocity increased above the 20% threshold of the gesture's maximum velocity for the movement away from the target (PVEL2) (Carignan et al., 2021; Oh et al., 2024). Movement offset (GOFFS) is then the point at which velocity reaches 20% of this second maximum velocity when moving away from the target.

[1] Younger speakers have larger adenoid glands, which tend to protrude beyond the edge of the pharyngeal wall. Inclusion of large portions of the adenoid glands obfuscated the variation in velum action captured by PC1.

[2] https://github.com/reedblaylock/VocalTract-ROI-Toolbox

An example of this can be seen in Figure 5 showing the example target phrase "bow no" with the corresponding spectrogram (top) and velum (middle) and TT (bottom) pixel intensity timeseries with indicated gestural landmarks.

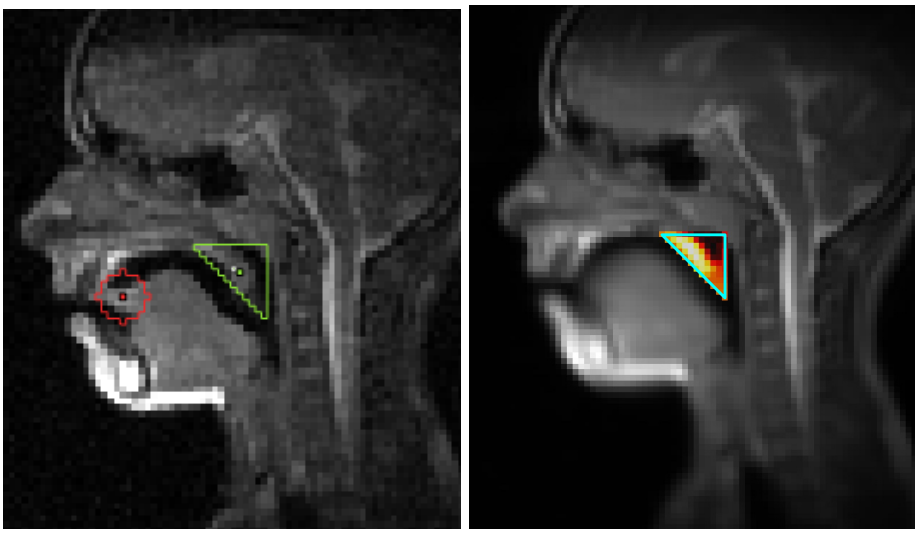

Figure 4: Example ROIs for the TT and Velum for spk8. The TT region tracks movement using average pixel intensity, while the velum region uses centroid tracking.

2.5 Measurements

The dependent variables calculated from the gestural landmarks can be found in Table 3, grouped by their relevance to timing, magnitude, and duration.

2.6 Statistics

For each variable, a number of statistical models were fit on the resulting data. To test overall effects of certain factors for each variable, linear-mixed models were fit using the `lmer` package in R. These models included fixed effects of syllable position, age, and frame, with interactions between age and syllable position, and random speaker effects and random speaker slopes. The primary focus is on fixed effects for the interaction between syllable position and age, as we predict developmental differences in the dependent variables as a function of the syllable position of the target nasals. Main effects from ANOVAs performed on the linear models using the `car` package are also reported when relevant. To test the difference between syllable positions within speaker, pairwise Wilcoxon tests were performed on all pairwise comparisons, e.g. onset versus coda. Following Shaw et al. (2021) and Oh et al. (2024), we use linear regression to investigate the stability of timing patterns as a function of variation in velum magnitude and duration. To control for collinearity between velum magnitude and duration in these models, we compute partial correlations using the residuals from linear regressions predicting magnitude and onset lag from duration. All statistical analysis was done on the raw values without normalizing. This follows from the values for each variable falling roughly on the same scale across speakers and this allows for interpreting model coefficients with respect to meaningful real-world values. Full statistical results are reported in the Appendix.

3 Results

3.1 Timing

To provide a sense of the distributions for onset lag across syllable positions, Figure 6 shows per speaker density plots for this variable, ordered by age. Negative lag here indicates early velum

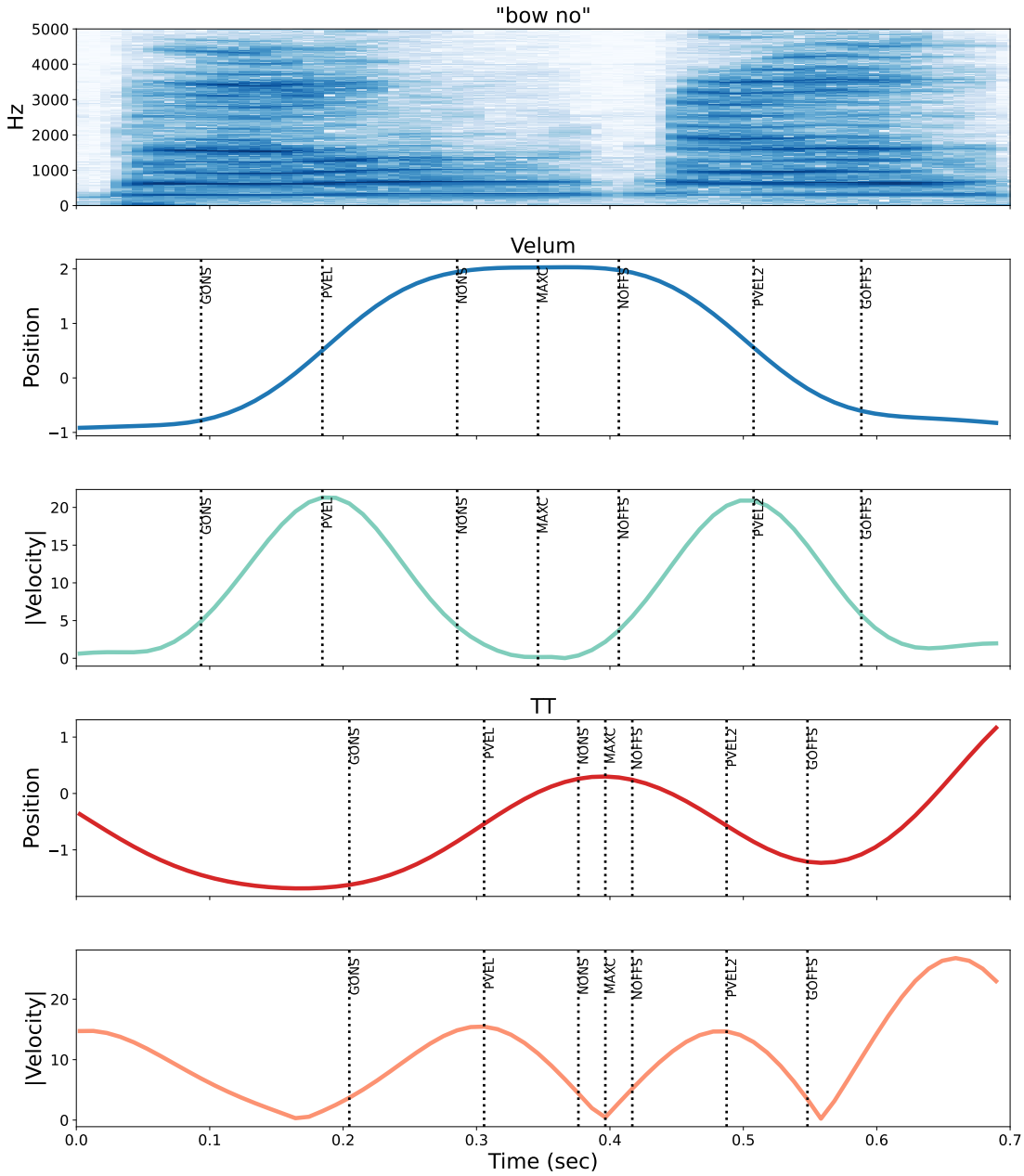

Figure 5: Example spectrogram (top) and TT (middle) and velum (bottom) timeseries from the ROI analysis with gestural landmarks indicated.

		Measure
Timing (ms)	ONSET LAG	Velum ONS - TT ONS
	TT PHASE	$\frac{\text{TT ONS - Velum ONS}}{\text{Velum Duration}}$
Magnitude		MAXC - ONS
Duration (ms)		OFFS - ONS

Table 3: Dependent variables used in the current study grouped by timing, displacement and duration.

lowering, while zero is indicative of sychrony between the velum and TT. A number of observations can be made here. First, the two oldest speakers, age 15, exhibit clear bimodal distributions that follow previous results from adults. Onset nasals tend to involve near synchrony between the timing of the velum and TT, while coda and geminate nasals tend to have early velum lowering. For all other speakers, there is a clear lack of bimodality in the distrbiutions, with early velum lowering being prevalent in all syllable conditions. The mixed model shows a significant interaction between onset lag and age for nasals in onset position ($\beta = 15.2, p < 0.001$), suggesting as speakers get older onset lag moves closer to zero. There is also a significant effect of the labial frame ($\beta = 34.8, p < 0.001$), indicating more synchronous timing in this frame than the alveolar frame.

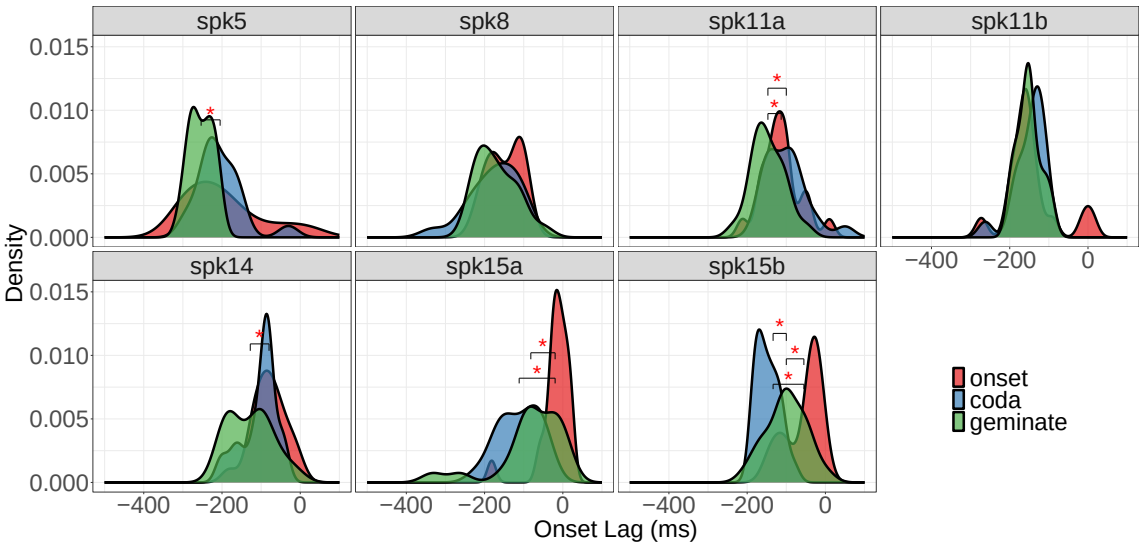

Figure 6: Density plots for onset lag compared across nasal syllable position with one plot per speaker. Speakers are arranged from top left to bottom right by age. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

While the results from onset lag indicate that the younger speakers employ early velum lowering regardless of the syllable position of the nasal, this variable does not reveal when these speakers time the onset of the TT relative to the velum gesture. To investigate this, we quantify at what relative point during the velum gesture interval that the TT gesture begins, termed here as TT phase. Per speaker density plots for this variable are shown in Figure 7. As a reference point, we also include the average relative time point for all of the velum gestural landmarks. As with onset lag, the two oldest speakers exhibit a more bimodal pattern where TT phase in onset nasals diverges from the other two conditions, with the TT being initiated closer to the onset of the velum gesture. The younger speakers all have largely unimodal distributions centered close to the time the velum reached peak velocity or its target. While the reference points are based on averages, the results here still suggest that the younger speakers wait until the velum is at or near its target

before initiating the TT gesture.

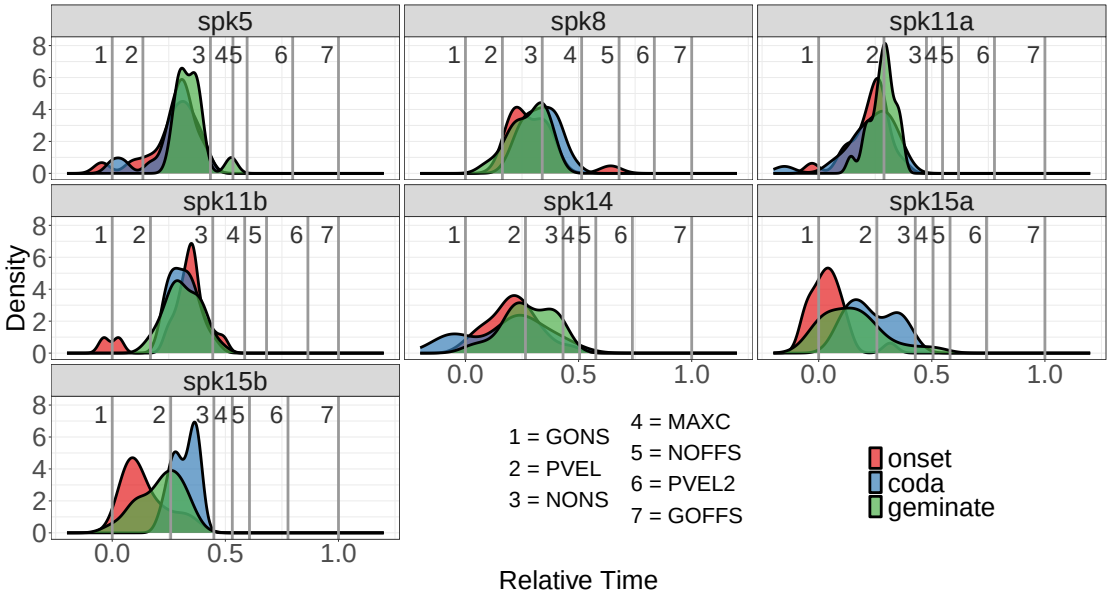

Figure 7: Density plots of the TT phase variable as compared across nasal syllable position with one plot per speaker. Speakers are arranged from top left to bottom right by age. The x-axis is relative time within the velum gesture duration and horizontal lines show average relative gestural landmark position for the velum.

3.2 Magnitude

Per speaker boxplots for velum magnitude are shown in Figure 8. As with onset lag, the two oldest speakers exhibit a clear distinction between onset nasals and both coda and geminate nasals, with onset nasals having significantly less velum magnitude. Interestingly, coda nasals also show much greater velum magnitude than geminates for these two speakers, likely attributable to coda effects. Some younger speakers produce greater velum magnitude in geminates than onsets, e.g. spk14 and spk11a and to a lesser extent in spk5 and spk8, though none of these speakers demonstrate the expected adult onset-coda distinction along this dimension. Results from the mixed model for velum magnitude show a significant interaction between age and onset position (($\beta = -0.08, p < 0.05$)), indicating less velum magnitude in onset nasals as one gets older. There was also a significant fixed effect of the labial frame ($\beta = 0.22, p < 0.001$), suggesting greater velum magnitude when producing nasals in this frame. In the alveolar frame, the TT likely maintains a more raised position in anticipation of the upcoming nasal, resulting in greater change in TT position than in the labial frame condition.

3.3 Duration

Per-speaker boxplots for velum duration are shown in Figure 10. As with onset lag and velum magnitude, the two oldest speakers exhibit the expected adult pattern in that onset nasals have siginificantly shorter velum gestures than coda and geminate nasals. Two of the younger speakers, spk8 and spk14, produce longer velum gestures in geminates than onsets and codas. Surprisingly, spk11a displays shorter velum gestures in codas than the other two syllable positions. Other speakers have fairly uniform velum gesture durations across syllable positions. Results from the mixed model show a significant interaction between age and onset position ($\beta = -32.7, p < 0.001$), indicating shorter velum gestures in onset nasals as one ages. The fixed effect of the labial frame was also significant ($\beta = -22.1, p < 0.01$), suggestive of shorter velum gestures in this frame condition.

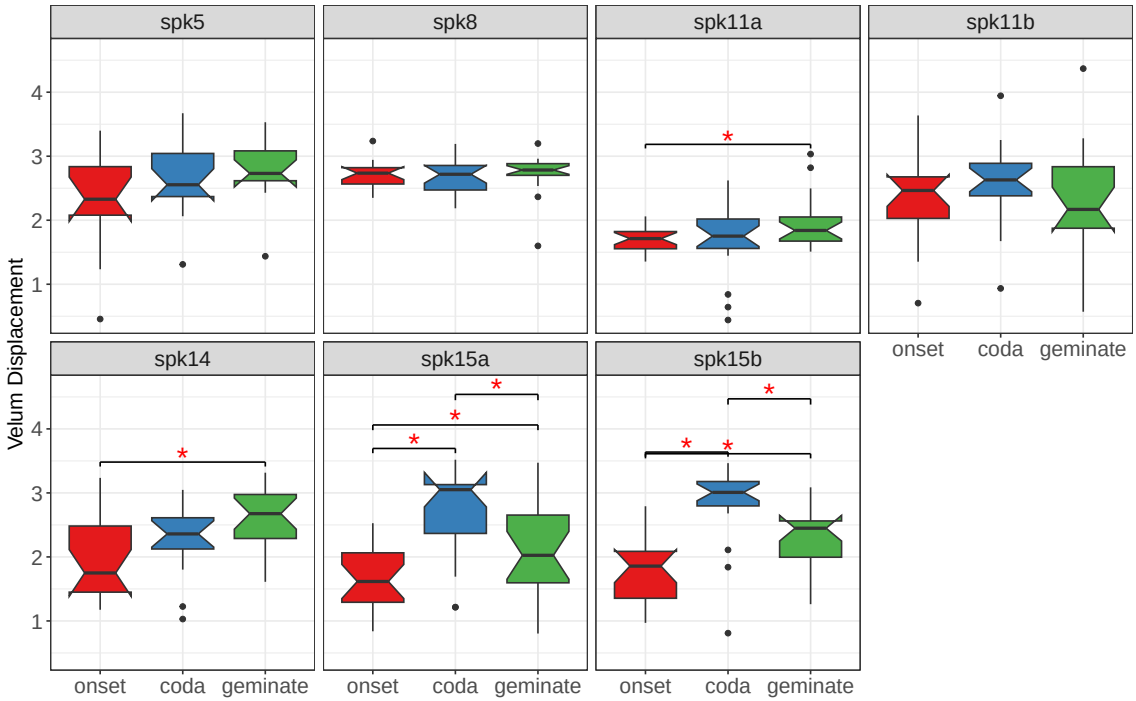

Figure 8: Per speaker box plots for velum displacement across syllable positions. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

The results for TT duration are shown in Figure 11. There is a general trend across speakers that geminate nasals have longer TT durations than both onsets and codas, though there is no difference between onset and coda nasals.

3.4 Timing stability

To investigate the stability of timing relationships between the velum and TT, we fit per-speaker linear regressions between timing variables and velum magnitude and duration. Significant correlations between the timing and magnitude or duration indicate a lack of stability, while absence of a correlation is suggestive of the timing relationship being stable for that speaker. In addition to onset lag, we also include velum target to TT onset lag (velum NONS - TT GONS, referred to as target-onset lag hereon), given that the results from TT phase showed that speakers tend to initiate the TT near the time the velum achieves its target. We also fit linear mixed models predicting each timing variable given velum magnitude and duration and their interaction with age (a three-way interaction).

As stated above, these regressions involve partial correlations to control for collinearity between velum magnitude and duration. To confirm this, we fit linear regressions predicting magnitude from duration for both the velum and TT. Figure 12 shows the results for this per-speaker. As suspected, a majority of speaker demonstrate a significant correlation between velum magnitude an duration. The same results for the TT are shown in Figure 13. All speakers except spk15b show a significant correlation between TT magnitude and duration. While these relationships are not fully consistent across speakers, controlling for this collinearity allows for consistency in our statistical models.

Figures 14 and 15 partial correlations between onset lag and velum magnitude and duration respectively. Noticeably, all speakers exhibit a significant correlation between onset lag and either velum magnitude or duration. While all speakers show this correlation for velum duration, a few speakers do not for velum magnitude. This consistency is highly indicative of a lack of stability

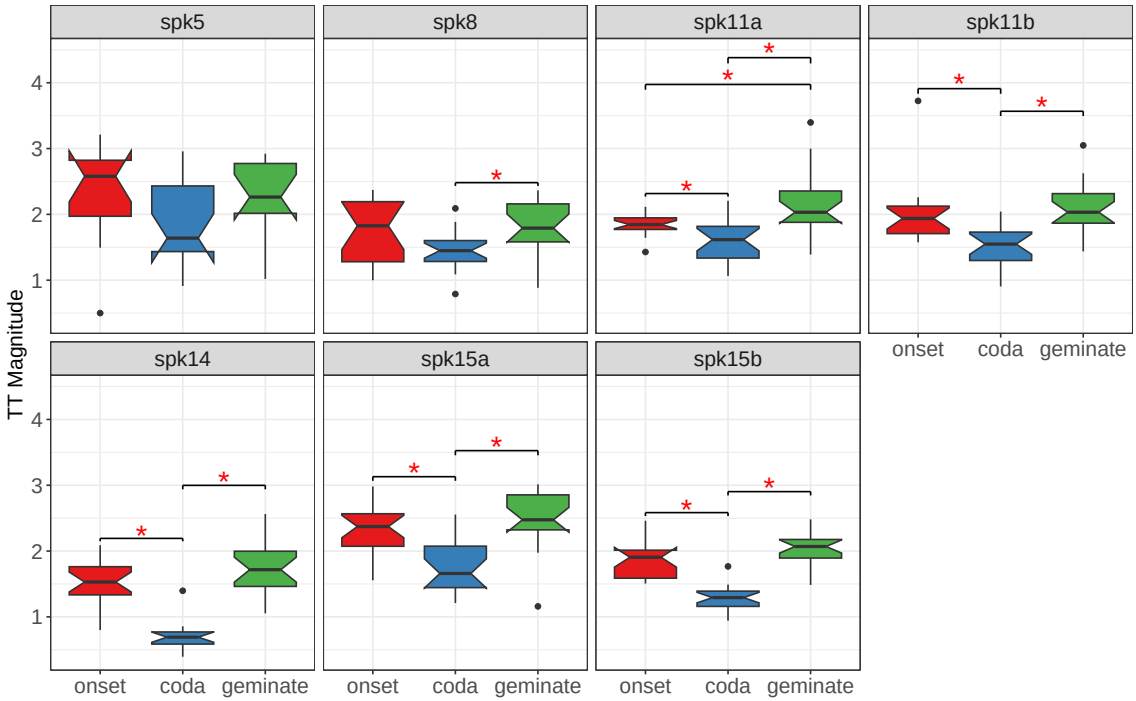

Figure 9: Per speaker box plots for TT magnitude across syllable positions. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

in this timing variable. The negative correlations across both variable point towards earlier velum lowering being accompanied by greater velum magnitude and longer velum gestures. The mixed model for onset lag shows main effects of velum duration ($F(1, 374.49) = 11.6$, $p < 0.001$), velum magnitude ($F(1, 372.5) = 8.5$, $p < 0.01$), the interaction between velum magnitude and duration ($F(1, 376.3) = 5.2$, $p < 0.05$), and frame ($F(1, 374.98) = 55.3$, $p < 0.001$). Age and interactions with age were not found to be significant.

Results from partial correlations between target-onset lag and velum magnitude and duration are shown in Figures 16 and 17 respectively. Note that negative target-onset lag indicates the velum target is achieved before the TT onset, while positive values mean the TT onset occurs first. With the exception of the two youngest speakers, spk5 and spk8, there is a clear trend across the older speakers that no significant correlations are found across the two sets of correlations. Spk5 shows significant correlations for both comparisons, while spk8 has a significant correlation in the duration comparison only. Compared to the results from onset lag, the target-onset relationship is much more stable across speakers. This closely mirrors results from Korean adults (Oh, 2021), though it has not been previously reported for English. The mixed model for target-onset lag shows significant main effects of velum duration ($F(1, 374.58) = 19.78$, $p < .001$), age ($F(1, 122.59) = 5.03$, $p = .027$), and frame ($F(1, 372.59) = 39.48$, $p < .001$). In addition, there are significant interactions between velum magnitude and duration ($F(1, 374.05) = 9.67$, $p = .002$), age and velum duration ($F(1, 374.67) = 13.95$, $p < .001$), and a significant three-way interaction between velum magnitude, duration, and age ($F(1, 373.77) = 7.10$, $p = .008$). Crucially, age and its interactions were not found to be significant in the model for onset lag, while both age and all of its interactions were significant in the target-onset lag model. We take this as evidence that, as suggested in the correlation plots, there are no developmental differences in the stability of onset lag, while for target-onset lag there are significant differences, with some younger speakers showing a lack of stability in this variable.

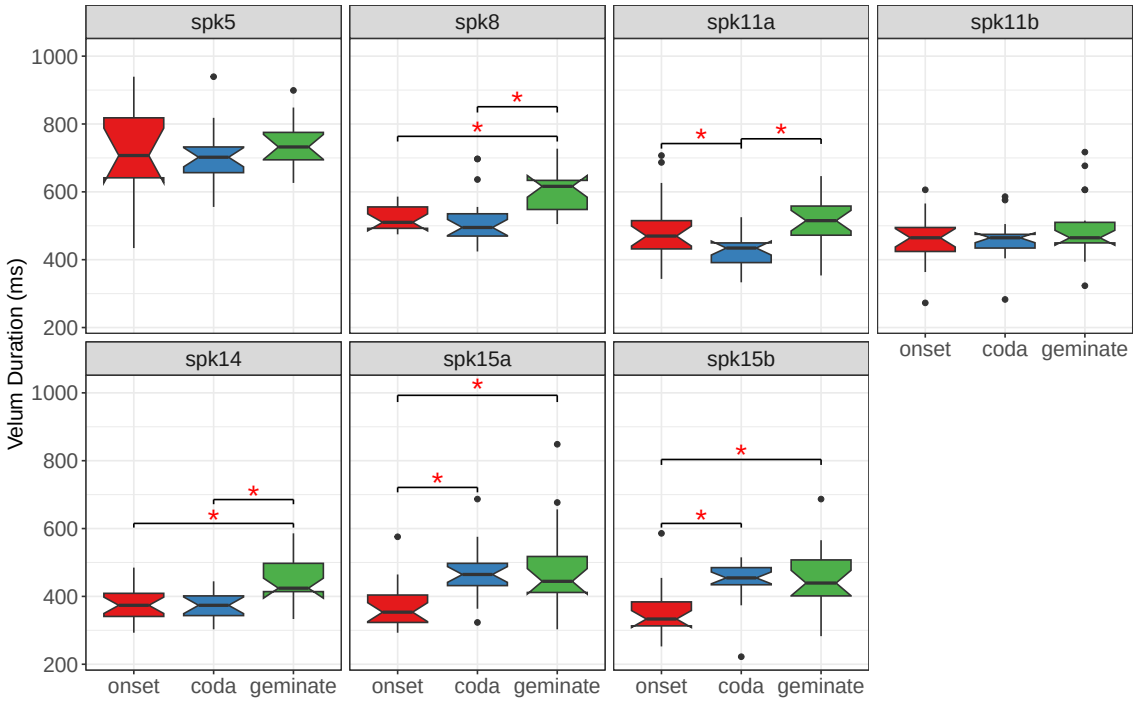

Figure 10: Per speaker box plots for velum gesture duration across syllable positions. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

4 Discussion

To recapitulate, the results showed that the two oldest speakers, age 15, essentially replicate the velum coordination patterns previously reported for adults. Both speakers exhibit near synchronous timing of the velum and TT gestures in onset nasals and produce early velum lowering in coda and geminate nasals. These speakers also modulate velum gesture duration and magnitude depending on syllable position, with onset nasals having lesser velum magnitude and shorter velum gestures than coda and geminate nasals. The younger speakers, ages 5-14, all employ early velum lowering, regardless of the syllable position of the nasal, and do not consistently make distinctions between onset and coda nasals in terms of velum magnitude and duration. For timing stability, across speakers it was found that velum target to TT onset lag was more stable than onset lag as a function of varying velum magnitude and duration. Some younger speakers, especially spk5, however, did not consistently display stability in either timing variable.

For timing, it was predicated that younger speakers will display a uniform timing strategy in velum coordination. This prediction was borne out as all speakers ages 5-14 exhibited largely unimodal distributions in onset lag and in the phasing of the TT. Regarding the timing strategy itself, we acknowledged two competing hypotheses, early velum lowering or synchronous timing, each with their own motivation. The results support the early velum lowering hypothesis as all younger speakers exhibited this strategy across all nasal syllable positions. This finding is in line with previous work showing that children have as much as or more anticipatory nasalization than adults (Flege, 1988; Leeper et al., 1998; Ha and Kuehn, 2006), given that this assumes early velum lowering. The synchronous timing hypothesis stemmed from work within the coupled oscillator framework that has proposed that in-phase or synchronous timing is more stable than anti-phase or asychronous timing and that in-phase coupling is more readily available to the child learner due to the frequency of CV syllables in early speech (Goldstein et al., 2006; Nam et al., 2009). Although the current study only focused on nasals, it nonetheless suggests that this reasoning does not extend

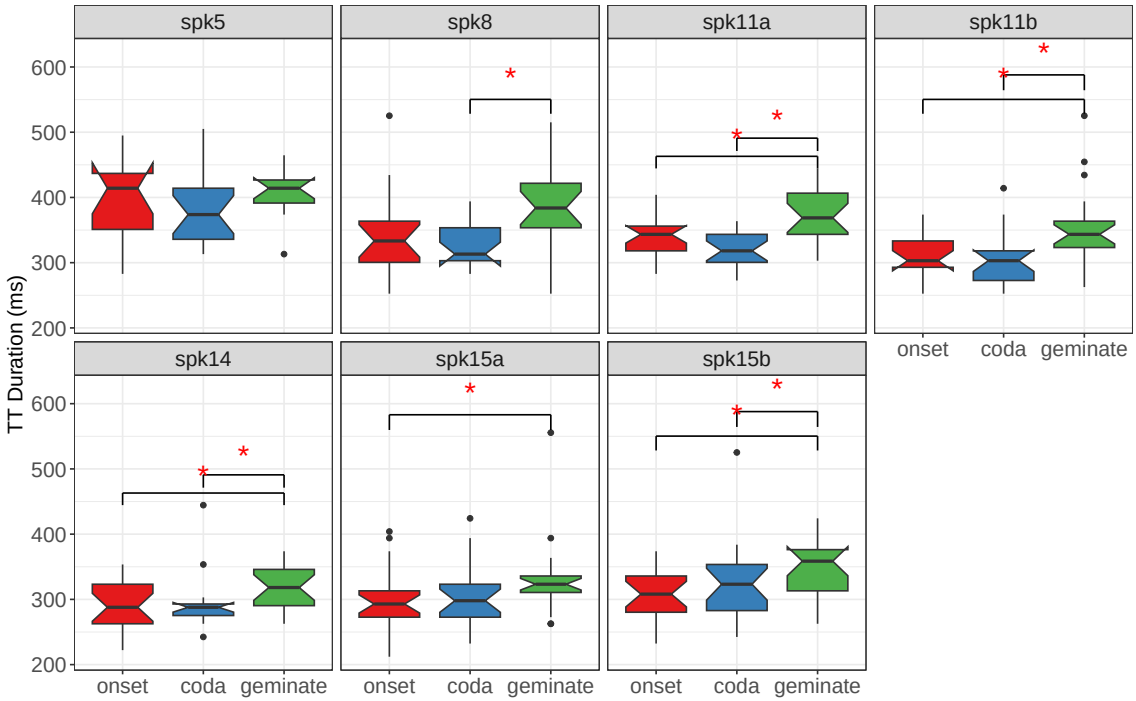

Figure 11: Per speaker box plots for TT gesture duration across syllable positions. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

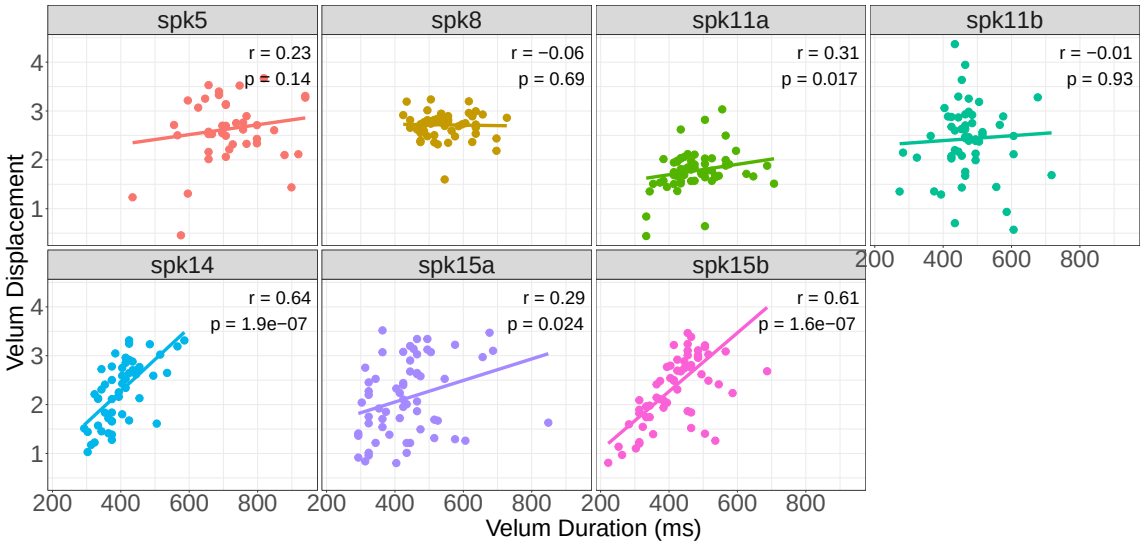

Figure 12: Per-speaker correlation graphs for velum displacement and duration with results from linear regression.

to the development of articulatory timing in complex segments.

The early velum lowering strategy is consistent with the view that velum lowering is likely more readily available to the child learner and to lower the velum early is easier than maintaining velum closure, which requires constant activation of the levator levi palatini, and timing its lowering to coincide with the TT raising gesture. Early infant vocalizations commonly feature velum lowering

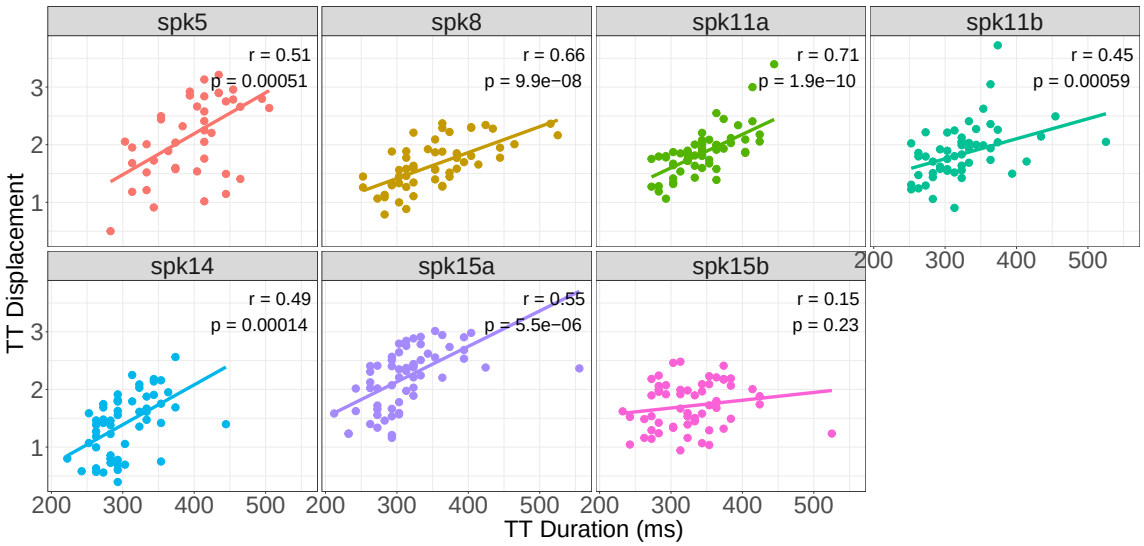

Figure 13: Per-speaker correlation graphs for TT displacement and duration with results from linear regression.

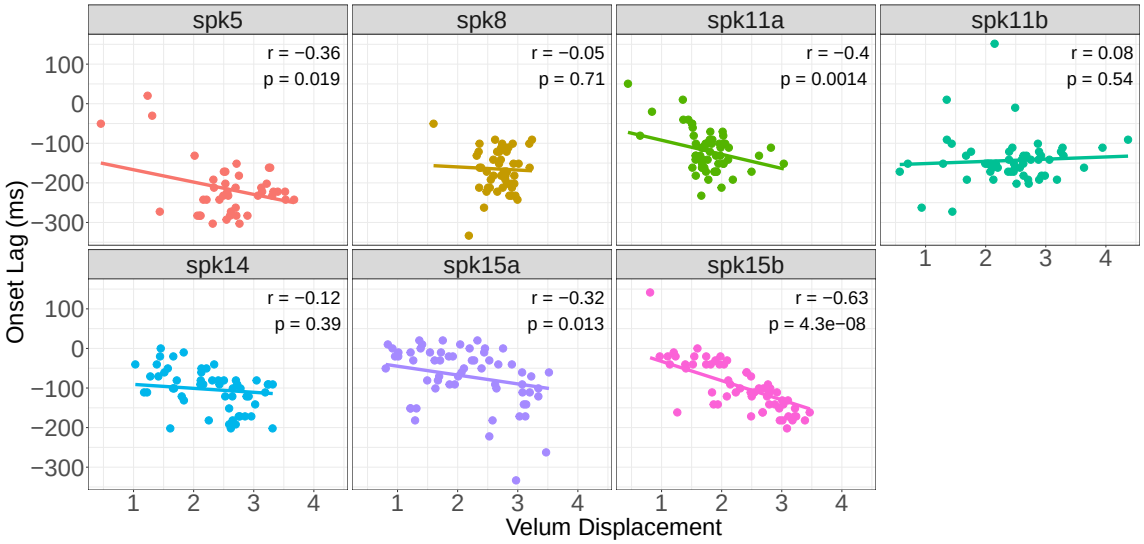

Figure 14: Per-speaker correlation graphs for onset lag and velum displacement with results from linear regression when controlling for duration.

and a lowered velum is the default state at rest (Zajac, 2015). The child must learn to raise the velum and maintain this closure during oral segments, an action that may be further complicated by growth and reduction of the adenoids between the ages of 4-15 (Zajac, 2015; Perry et al., 2017). This can be reframed as development in velum coordination entails learning to *inhibit* velum lowering in onset nasals and timing this event to closely coincide with the oral constriction. Although this explanation is largely a biomechanical one, language-specific phonology also plays a role here. We do not expect the same degree of velum lowering in similar contexts in languages that contrast oral and nasal vowels, e.g. French. To nasalize preceding vowels in English does not endanger any phonological contrasts, while in French it would. We speculate then that in a language such as French that children still lower the velum earlier than adults, but that velum lowering for nasal consonants is distinct in timing from that for nasal vowels. Children likely maintain this distinction

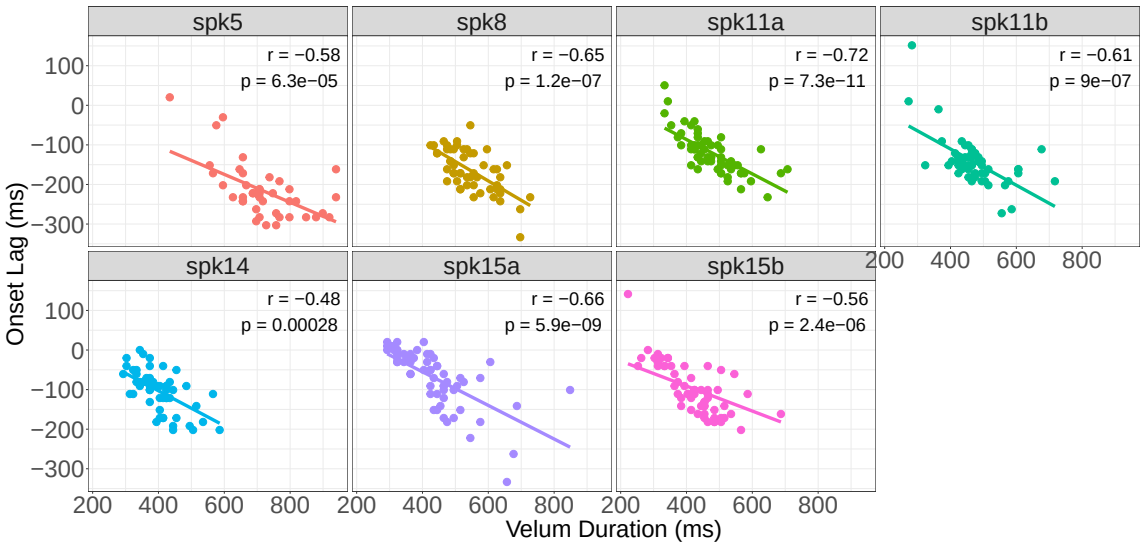

Figure 15: Per-speaker correlation graphs for onset lag and velum gesture duration with results from linear regression when controlling for displacement.

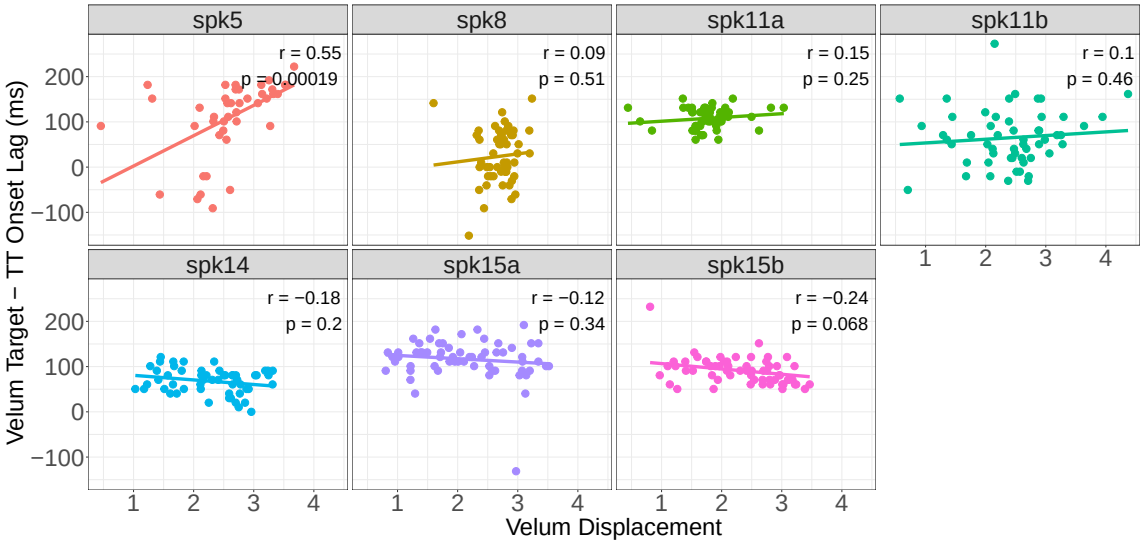

Figure 16: Per-speaker correlation graphs for velum target to TT onset lag and velum displacement with results from linear regression when controlling for duration.

from an early age, but velum lowering for nasal consonants is still earlier than it is in adults.

This explanation for early velum lowering also does not necessarily apply to the development of articulatory timing in other complex segments, though it is compatible with the explanation by Howson and Redford (2021) for the acquisition of articulatory timing in liquids. The authors postulate that tongue body constrictions for vowels are well-formed and readily available to the child learner and thus to form an early tongue body constriction in onset liquids is easier than inhibiting this action and timing it near synchronously with the oral constriction. Concretely, children prefer articulatory postures that are familiar, i.e. vowel-like tongue body constrictions and a lowered velum state, and are biased towards these familiar states or actions when they useful. This could also be framed as articulatory reuse, in that there is a preference to reuse articulatory primitives (Faytak, 2018), such as velum lowering or tongue body retraction, as an efficient meaning

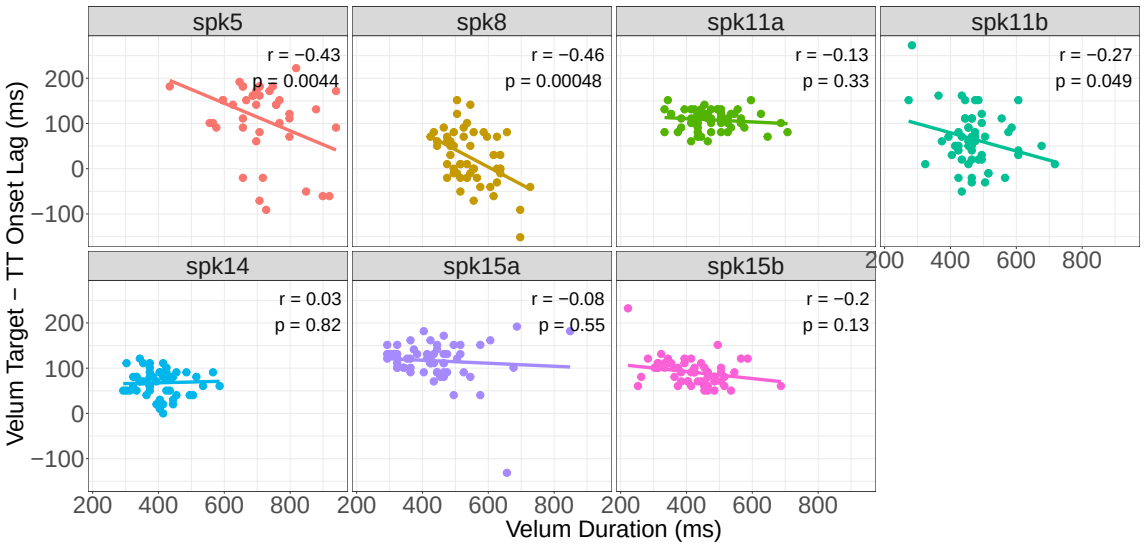

Figure 17: Per-speaker correlation graphs for velum target to TT onset lag and velum gesture duration with results from linear regression when controlling for displacement.

of achieving multiple constrictions for complex segments. Learning to inhibit the reused gesture then forms a critical component in the development of articulatory coordination in complex segments.

Investigation into the stability of timing between the velum and TT showed that, while all speakers displayed some variation in onset lag as a function of variation in velum magnitude or duration, many speakers consistently did not show this variation in the lag between velum target achievement and TT onset. This closely mirrors the results from Korean adults (Oh, 2021). As was suggested for Korean, we take these results to mean that the velum target to TT onset timing is encoded in the cognitive representations for nasals in American English. There was also the question of whether or not younger speakers are even coordinating the velum and the TT at all. Although speakers could exhibit a lack of distinction in velum-TT timing due to changes in syllable position, we suggested that stability in this timing relationship along some dimension could still indicate that there is in fact coordination between the two gestures. This can be seen clearly for spk11a and spk14, who both produce a fairly uniform timing strategy across syllable positions, but show stability in the velum target to TT onset lag when there is variation in velum magnitude and duration. The same cannot be said for spk5, spk8, and spk11b, who exhibit a lack of stability in both onset lag and velum target to TT onset lag, while also producing uniformly early velum lowering. This may suggests these speakers have yet to learn to coordinate the velum and TT.

Although the results from timing display developmental stages of unimodal to bimodal timing patterns, there is a clear lack of developmental progression from one stage to the other. All speakers aged 5-14 show fairly uniform distributions and then the two oldest speakers suddenly exhibit clear bimodality.

One potential confound for the lack of distinction in younger speakers between onset and coda nasals along the dimensions investigated here is that perhaps these speakers just simply aren't making any distinction among the different target phrases within the labial/coronal frames. That is, they aren't distinguishing "bow no" and "bone oh", but are rather producing a single disyllabic sequence across both sets of tokens. None of the other results, e.g. TT duration or magnitude, suggest a distinction between the onset and coda positions. To investigate this, we looked at the duration of the vowels in the target phrases, results for which are in Figure 18. The results indicate that for nearly all speakers the vowel in the second syllable is much longer in the coda condition than in the onset condition. This difference is significant for all speakers except spk8, though this distinction is still evident for this speaker. This findings makes sense given that the V2 in the coda condition appears in isolation. This provides evidence that, while the younger speakers are not making timing, magnitude, or duration distinctions in velum coordination dependent on varying

syllable positions, they are still actually producing distinct utterances across the onset and coda conditions.

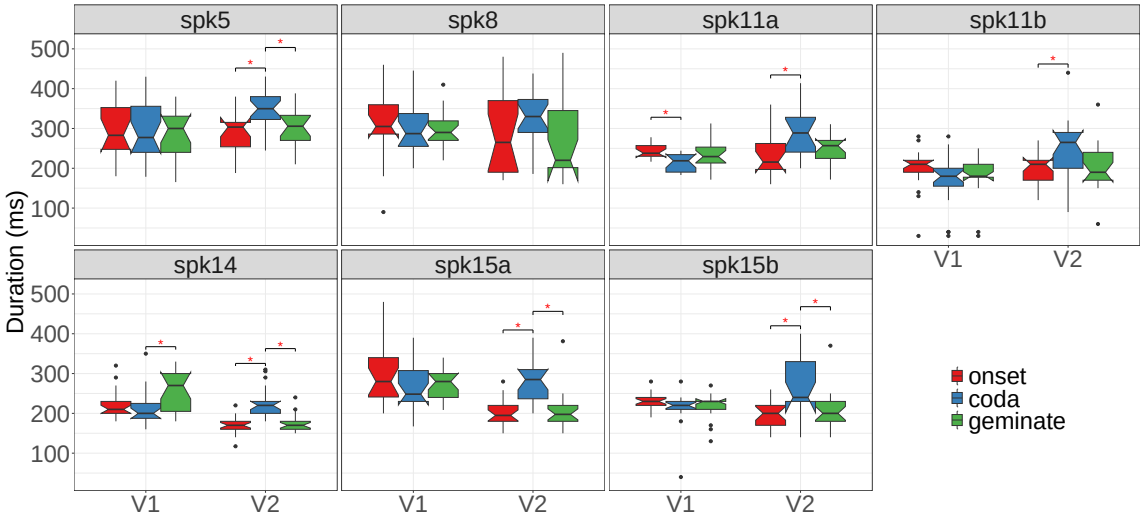

Figure 18: Per speaker box plots for vowel duration across syllable positions, where V1 and V2 indicate the vowel in the first and second syllable respectively. Red stars indicate significant results from per-speaker pairwise Wilcoxon tests on syllable position comparisons.

Various extension of the current study are feasible. First, the current study only investigated American English, which itself has specific velum timing patterns. Replicating this study on other languages with different velum timing patterns, e.g. Korean, French or German, is highly encouraged. Language-specific constraints on nasalization are naturally expected to impact developmental patterns. For example, given that French contrasts oral and nasal vowels, while American English does not, an early velum lowering strategy has different implications for French than it does American English. Cross-linguistic findings can also reveal aspects of the development of velum coordination that are due to physiological factors, such as the reduction in size of the adenoid glands potentially impacting velum closure. Second, the current study did not investigate coordination of the velum and TT with either adjacent vowel. It has been proposed that in onset nasals the TT, velum, and vowel are initiated sychronously (), while for coda nasals it is a bit unclear what is expected (). For a simplex segment such as /t/, the vowel and TT are expected to be phased asychronously, but when an additional gesture is present, as in nasals, how such an additional gesture is timed with respect to the vowel is largely unknown.

There are a number of limitatons to the current study. First, the participants were all female, which limits generalizing the results of the study to all children, especially given developmental differences between boys and girls (Smith and Zelaznik, 2004). Replicating the current study with male subjects would allow for better generalizing the results to speech development more broadly. If boys ages 5-14 also exhibit the patterns shown here with respect to timing, magnitude, and duration of the velum, it would be more suggestive of these patterns being typical in the development of American English nasals. Second, the sample size here is relatively small, also limiting generalizability, especially given that developmental trajectories can vary highly across people (Nittrouer, 1993). As noted above, there is a clear lack of progression from the unimodal timing patterns in the younger subjects to the bimodal patterns of the two oldest speakers. Whether or not this actually reflects the typical developmental trajectory of velum timing or is due to the small sample size can only be determined by replicating this study on more subjects. Lastly, the current study only included nasals in the low vowel context. Although this was intentional, it does not allow us to account for potential velum lowering effects intrinsic to the vowels themselves (Bell-Berti and Krakow, 1991). It has been established that low vowels have more velum lowering than high vowels (Bell-Berti, 1993; Zajac, 2015) and, as discussed above, we cannot rule out the

role of these effects in biasing the younger subjects towards early velum lowering.

5 Conclusion

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Appendix

A: Results from statistical tests

Table 4: Fixed-effect coefficients for the model comparing onset lag across syllable position, age, frame. Formula: lag_onset ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	-297.403	29.782	6.185	-9.986	4.83e-05
condition:coda	44.995	47.185	5.426	0.954	0.38083
condition:geminate	-17.294	28.257	15.886	-0.612	0.54917
age	15.236	2.478	5.781	6.148	0.00098
frame:labial	34.786	5.581	370.378	6.233	1.24e-09
condition:coda:age	-6.737	3.971	5.304	-1.697	0.14719
condition:geminate:age	-1.821	2.327	14.290	-0.783	0.44646

Table 5: Fixed-effect coefficients for the model comparing TT phase across syllable position, age, frame. Formula: TT phase ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	0.433143	0.093113	5.295	4.652	0.00482
condition:coda	-0.097176	0.142864	5.170	-0.680	0.52564
condition:geminate	-0.010571	0.075514	10.061	-0.140	0.89143
age	-0.018412	0.007828	5.152	-2.352	0.06388
frame:labial	-0.022247	0.014735	390.193	-1.510	0.13189
condition:coda:age	0.010992	0.012066	5.124	0.911	0.40311
condition:geminate:age	0.005488	0.006318	9.608	0.869	0.40624

Table 6: Fixed-effect coefficients for the model comparing velum displacement across syllable position, age, frame. Formula: displacement ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	2.86806	0.36789	5.517	7.796	0.000352
condition:coda	-0.50913	0.48520	5.289	-1.049	0.339586
condition:geminate	-0.04327	0.28957	5.516	-0.149	0.886526
age	-0.08091	0.03078	5.268	-2.629	0.044288
frame:labial	0.22426	0.05710	363.570	3.927	0.000103
condition:coda:age	0.08115	0.04084	5.174	1.987	0.101742
condition:geminate:age	0.03254	0.02383	4.971	1.365	0.230692

Table 7: Fixed-effect coefficients for the model comparing TT magnitude across syllable position, age, frame. Formula: TT magnitude ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	2.06301	0.40847	5.252	5.051	0.00342
condition:coda	-0.19010	0.25251	5.390	-0.753	0.48311
condition:geminate	-0.07592	0.19252	21.468	-0.394	0.69720
age	-0.02104	0.03442	5.160	-0.611	0.56696
frame:labial	0.23653	0.03912	369.778	6.046	3.62e-09
condition:coda:age	-0.02691	0.02117	5.183	-1.271	0.25766
condition:geminate:age	0.01932	0.01582	19.229	1.221	0.23681

Table 8: Fixed-effect coefficients for the model comparing velum gesture duration across syllable position, age, and frame. Formula: dur_velum ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	850.242	44.057	5.244	19.298	4.51e-06
condition:coda	-98.614	67.941	5.452	-1.451	0.20166
condition:geminate	-11.254	44.793	9.845	-0.251	0.80680
age	-32.696	3.668	4.913	-8.913	0.000324
frame:labial	-22.137	8.095	369.333	-2.735	0.006546
condition:coda:age	10.106	5.718	5.328	1.767	0.13379
condition:geminate:age	6.498	3.704	9.006	1.754	0.11323

Table 9: Fixed-effect coefficients for the model comparing TT gesture duration across syllable position, age, and frame. Formula: dur_TT ∼ condition * age + frame + (1 + condition | speaker)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	411.641	21.589	6.810	19.067	3.66e-07
condition:coda	-44.223	21.061	14.858	-2.100	0.05326
condition:geminate	18.608	19.892	240.042	0.935	0.35049
age	-9.404	1.793	6.314	-5.246	0.00165
frame:labial	45.098	4.288	374.147	10.518	¡2e-16
condition:coda:age	3.268	1.753	13.882	1.864	0.08357
condition:geminate:age	1.367	1.629	227.864	0.840	0.40205

Table 10: Fixed-effect coefficients for the model predicting onset lag: lag_onset ∼ mag_velum * age * dur_velum + frame + (1 — spk_label)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	316.100	131.000	332.322	2.412	0.016401
mag_velum	-151.200	52.010	372.500	-2.908	0.003861
age	-10.450	9.948	318.600	-1.051	0.294247
dur_velum	-0.7193	0.2109	374.500	-3.411	0.000717
frame:labial	32.800	4.412	375.000	7.435	7.17e-13
mag_velum:age	7.248	4.058	374.200	1.786	0.074915
mag_velum:dur_velum	0.1892	0.0826	376.300	2.290	0.022576
age:dur_velum	0.01300	0.01723	374.000	0.755	0.450906
mag_velum:age:dur_velum	-0.009457	0.007034	376.800	-1.344	0.179627

Table 11: Type III ANOVA (Satterthwaite) for the model: lag_onset ∼ mag_velum * age * dur_velum + frame + (1 — spk_label)

Effect	Sum Sq	Mean Sq	NumDF	DenDF	F value	p value
mag_velum	14198	14198	1	372.50	8.4539	0.0038606
age	1854	1854	1	318.57	1.1037	0.2942466
dur_velum	19542	19542	1	374.49	11.6360	0.0007174
frame	92835	92835	1	374.98	55.2775	7.174e-13
mag_velum:age	5357	5357	1	374.20	3.1897	0.0749152
mag_velum:dur_velum	8807	8807	1	376.30	5.2439	0.0225764
age:dur_velum	957	957	1	374.02	0.5696	0.4509059
mag_velum:age:dur_velum	3035	3035	1	376.82	1.8074	0.1796269

Table 12: Fixed-effect coefficients for the model predicting tarons lag: lag_tarons ∼ mag_velum * age * dur_velum + frame + (1 — spk_label)

Effect	Estimate	Std. Error	df	t value	p value
(Intercept)	463.773827	146.106017	157.715204	3.174	0.001807
mag_velum	-85.295657	54.450146	374.791069	-1.566	0.118078
age	-25.239466	11.250832	122.586522	-2.243	0.026670
dur_velum	-0.980639	0.220490	374.581172	-4.448	1.15e-05
frame:labial	28.767703	4.578421	372.592324	6.283	9.26e-10
mag_velum:age	5.554326	4.243580	374.540869	1.309	0.191378
mag_velum:dur_velum	0.268118	0.086206	374.047811	3.110	0.002013
age:dur_velum	0.067294	0.018019	374.673098	3.735	0.000217
mag_velum:age:dur_velum	-0.019542	0.007332	373.773839	-2.665	0.008028

Table 13: Type III ANOVA (Satterthwaite) for the model: lag_tarons ∼ mag_velum * age * dur_velum + frame + (1 — spk_label)

Effect	Sum Sq	Mean Sq	NumDF	DenDF	F value	p value
mag_velum	4420	4420	1	374.79	2.4539	0.1180776
age	9064	9064	1	122.59	5.0326	0.0266705
dur_velum	35628	35628	1	374.58	19.7806	1.146e-05
frame	71109	71109	1	372.59	39.4802	9.257e-10
mag_velum:age	3086	3086	1	374.54	1.7132	0.1913784
mag_velum:dur_velum	17423	17423	1	374.05	9.6734	0.0020130
age:dur_velum	25120	25120	1	374.67	13.9470	0.0002173
mag_velum:age:dur_velum	12794	12794	1	373.77	7.1033	0.0080277